

CBSE Class 10 Mathematics — Probability

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains *questions. All questions are compulsory.*
1. Questions are grouped into sections A through E.
2. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. If the probability of an event E occurring is 0.65, what is the probability of the event 'not E '?
(A) (A) 0.25
(B) (B) 0.35
(C) (C) 0.45
(D) (D) 0.65
2. The probability of winning a game is $\frac{7}{12}$. What is the probability of losing the game?
(A) (A) $\frac{1}{12}$
(B) (B) $\frac{4}{12}$
(C) (C) $\frac{5}{12}$
(D) (D) $\frac{7}{12}$
3. A bag contains balls of different colours. If the probability of picking a red ball is $\frac{2}{9}$, what is the probability that the ball picked is NOT red?
(A) (A) $\frac{1}{9}$
(B) (B) $\frac{2}{9}$
(C) (C) $\frac{5}{9}$
(D) (D) $\frac{7}{9}$
4. If $P(E) = 0.07$, find $P(\text{not } E)$.
(A) (A) 0.93
(B) (B) 0.97
(C) (C) 0.03
(D) (D) 0.07
5. A student solves a problem correctly with a probability of 0.85. What is the probability that the student solves the problem incorrectly?

- (A) (A) 0.05
 (B) (B) 0.10
 (C) (C) 0.15
 (D) (D) 0.25
6. Assertion (A): If the probability of winning a game is 0.7, then the probability of losing the game is 0.3. Reason (R): The sum of the probabilities of all complementary events is 1.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
7. Assertion (A): The probability of an event happening is 0.45, then the probability of the event not happening is 0.65. Reason (R): For any event E, $P(\bar{E}) = 1 - P(E)$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
8. Assertion (A): If a fair coin is tossed, the probability of getting a head is 0.5 and the probability of not getting a head is 0.5. Reason (R): Getting a head and not getting a head are complementary events.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
9. Assertion (A): If the probability of selecting a red ball from a bag is $\frac{2}{7}$, the probability of not selecting a red ball is $\frac{5}{7}$. Reason (R): The probability of an impossible event is 1.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false

- (D) (D) Assertion (A) is false but Reason (R) is true
10. Assertion (A): If $P(E) = 0.12$, then $P(\bar{E}) = 0.88$. Reason (R): $P(E) + P(\bar{E}) = 1$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section D: Long Answer (4-5 marks each)

1. A box contains 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, find the probability that it bears (i) a two-digit number, (ii) a perfect square number, (iii) a number divisible by 5. Use the concept of complementary events to verify the probability of the event 'not getting a perfect square'.
2. A game consists of tossing a one-rupee coin 3 times and noting its outcome each time. Hanif wins if all the tosses give the same result i.e., three heads or three tails, and loses otherwise. Calculate the probability that Hanif will lose the game using the complementary event property.
3. A bag contains 5 red balls and some blue balls. If the probability of drawing a blue ball is double that of a red ball, find the number of blue balls in the bag. Further, if a ball is drawn at random, what is the probability of not drawing a red ball?
4. Two dice are thrown simultaneously. Find the probability of getting (i) a sum of 8, (ii) a doublet, (iii) a sum greater than 9. Use the complementary event concept to find the probability of getting a sum not equal to 8.